WILEY

Society for Research in Child Development

Maternal Separation in Bonnet Monkey Infants: Altered Attachment and Social Support

Author(s): Martin Reite, Kristine Kaemingk and Maria L. Boccia

Source: *Child Development*, Apr., 1989, Vol. 60, No. 2 (Apr., 1989), pp. 473-480

Published by: Wiley on behalf of the Society for Research in Child Development

Stable URL: http://www.jstor.com/stable/1130991

REFERENCES
Linked references are available on JSTOR for this article:
http://www.jstor.com/stable/1130991?seq=1&cid=pdf-reference#references_tab_contents
You may need to log in to JSTOR to access the linked references.

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

Wiley and *Society for Research in Child Development* are collaborating with JSTOR to digitize, preserve and extend access to *Child Development*

JSTOR

This content downloaded from
81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms

Maternal Separation in Bonnet Monkey Infants: Altered Attachment and Social Support

Martin Reite, Kristine Kaemingk, and Maria L. Boccia

University of Colorado Health Sciences Center

REITE, MARTIN; KAEMINGK, KRISTINE; and BOCCIA, MARIA L. *Maternal Separation in Bonnet Monkey Infants: Altered Attachment and Social Support.* CHILD DEVELOPMENT, 1989, **60**, 473–480. Maternal separation in young pigtail (*M. nemestrina*) monkeys results in an agitation-depression reaction that is a useful animal model of the behavioral and physiological consequences of separation and loss experiences in humans. In this study, 5 social-group-living bonnet (*M. radiata*) monkey infants were separated from their mothers for 4 days, and behavioral, physiological, and sleep-pattern changes were monitored. Behavioral and physiological changes were consistent with an agitation reaction. There was evidence of depression in both behavioral and physiological measures; these changes were not as pronounced as in pigtail infants, however. All infants were adopted by another adult female during separation. Our findings suggest that, in bonnet monkey infants, adoption by a female within the social group ameliorates the response to loss.

Loss of an attachment object may be a potent pathogen in terms of its ability to precipitate psychological symptoms, especially depression, and health-related morbidity in vulnerable individuals (Brown & Harris, 1978; Dohrenwend & Dohrenwend, 1974; Laudenslager & Reite, 1984; Myers, Lindenthal, & Pepper, 1972; Paykel et al., 1969). The likelihood of significant depressive symptoms in response to loss appears to be lessened by the availability of social support (Brown, Andrews, Harris, Adler, & Bridge, 1986; Cassel, 1976; Cobb, 1976; Monroe, Bromet, Connell, & Steiner, 1986), although the nature and preconditions for the effectiveness of such social support have not yet been clearly delineated.

Nonhuman primate separation studies represent valuable animal model systems of human responses to loss and associated moderating factors. Such studies have demonstrated potent behavioral and physiological sequelae to separation from or loss of an attachment object (Kalin & Carnes, 1984; Reite & Short, 1983). Research with several species of monkeys, including rhesus (*M. mulatta*) and pigtails (*M. nemestrina*), has demonstrated a two-stage "protest-despair" or "agitation-depression" behavioral reaction in infants temporarily separated from their mothers (Hinde, Spencer-Booth, & Bruce, 1966; Kaufman & Rosenblum, 1967; Seay, Hansen, & Harlow, 1962). In infants of these species, the initial period of agitation is characterized by vigorous protests with much "cooing" vocalization and increases in activity as the separated infant apparently searches for the lost mother. The subsequent period of behavioral "despair" or "depression" is characterized by decreased social play, increased attempts to make contact with other adult females, and development of a characteristic sad facial expression and "slouched" posture. This behavioral response to separation from the mother is phenomenologically strikingly similar to the protest-despair response described in human children who have been separated from their mothers (Bakwin, 1949; Bowlby, 1960; Spitz, 1946).

In pigtail monkey infants, the separation-induced agitation-depression reaction is accompanied by pronounced disturbances in the regulation of heart rate (HR) and rhythm (Reite, Short, Kaufman, Stynes, & Pauley, 1978; Seiler, Cullen, Zimmerman, & Reite, 1979), body temperature (BT) (Reite et al., 1978), HR and BT circadian rhythms, and nocturnal sleep patterns (Reite, Seiler, Crowley, Hydinger-MacDonald, & Short, 1982; Reite & Short, 1978). It has been hypothesized that these changes are due, at least in part, to the disruption of the social bond between the mother and infant, although other

This research was supported by USPHS grant no. MH19514 to M. Reite. M. Reite is supported by NIMH Research Scientist Award no. 5K02 MH46335. Animal subjects in this study were treated in accordance with USPHS Policy on Humane Care and Use of Laboratory Animals. This research was approved by the UCHSC Institutional Animal Care and Use Committee.

[*Child Development*, 1989, **60**, 473–480. © 1989 by the Society for Research in Child Development, Inc. All rights reserved. 0009-3920/89/6002-0013$01.00]

This content downloaded from 81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms

influences, such as alterations in nutritional status, may be involved (Reite & Capitanio, 1985; Reite, Short, Seiler, & Pauley, 1981).

Evidence that social environment may play a role in modulating the response to separation or loss comes from several sources. First, in contrast to the pronounced agitation-depression reaction seen following maternal separation in pigtail infants, bonnet infants, a closely related species, exhibit less behavioral change following maternal separation in a social group environment (Rosenblum & Kaufman, 1968). Bonnet infants are raised in a social environment that differs from that of pigtail infants. Bonnet females exhibit more "aunting" behaviors, and infants therefore spend extensive time with other adult females (Reite & Short, 1986; Rosenblum & Kaufman, 1967). Second, in the mother-infant separation paradigm, Caine and Reite (1981) found that those separated pigtail infants who had greater peer contact in the social group experienced a less pronounced decrease in heart rate in response to loss of the mother.

The present study was designed to quantify and evaluate the physiological and behavioral response to maternal separation in bonnet infants in conspecific social groups. The hypothesis was that physiological responses to maternal separation in bonnet infants should be qualitatively similar to but quantitatively less than the response in pigtail infants, and, further, that bonnet infants would likely have more physical contact with other adult females during the separation period.

Method

Subjects

The subjects were five mother-reared bonnet infants (three males, two females). The animals lived in social groups containing one adult male and six to 10 adult females, some with infants. The groups were housed in pens measuring 2.1 × 2.5 × 4.0 m with glazed cinderblock walls, wire mesh ceilings, and plastic pipes positioned on one or more walls. One or two walls contained one-way windows for observational purposes. The groups were fed monkey biscuits and fruit at 0900 hours daily, and water was available ad lib. Timer-controlled fluorescent lights were on between 0700 and 2000 hours. The groups had minimal human contact other than that required for weekly pen cleaning, biannual TB testing, and occasional visits to the veterinarian.

Mother-infant separations occurred when the infants were a mean age of 173 ± 36 (SD) days. Following 4 (24-hour) days of preseparation baseline, the mother was removed from the social group for 4 days. The infant remained in the familiar social environment for the entire separation period.

Physiological Data Collection

At a mean age of 144 ± 37 (SD) days, the animals were surgically implanted under general (sodium pentobarbital) anesthesia with seven-channel PAM/FM hybrid biotelemetry units that transmitted EOG, EMG, EKG, body temperature, and three channels of EEG (Pauley & Reite, 1981; Reite, Walker, & Pauley, 1973). The units contained a bistable magnetically activated latching reed switch and could be switched on and off externally. Following recovery from the surgical procedure, the telemetry units were turned on and physiological data recording began.

When the units were on, data were transmitted 24 hours a day and were processed on-line by a dedicated LSI/11-03 microprocessor (Reite & Short, 1983). The on-line computer system sampled HR and BT every minute. HR samples were based on 20-sec EKG epochs each minute. In addition, a research assistant took a 20-sec HR and BT sample every 15 min to confirm computer accuracy, and made a notation of the infant's activity at the time. At night, this behavioral sample (obtained by means of infrared video) included with whom the infant was sleeping.

Physiological Variables

Physiological variables included measures of mean day and mean night HR and BT, estimates of HR and BT circadian rhythms, and measures of nocturnal sleep variables. Mean day HR was computed by averaging all HR samples obtained from 1000 to 1600 hours; mean night HR was the mean of all HR samples obtained between 2200 and 0400 hours. The same time epochs were used in computing mean day BT and mean night BT.

To assess the circadian component of HR and BT, a 24-hour cosine was fit using a least-squares technique to the computer-collected HR and BT samples from 0900 one day to approximately 0900 the next. Cosine fitting is one method of extracting circadian rhythm information from an otherwise complex waveform (Halberg, Tong, & Johnson, 1967; Reite & Short, 1981). Circadian variables obtained included the level, amplitude, and phase of the best-fit cosine curve. The cosine level approximates the mean of the 24-hour data. The amplitude represents the maximum deviation of the best-fit cosine above or below

This content downloaded from
81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms

the level (about half the maximum day-night difference). The phase represents the time (in hours) from 0900 to the peak amplitude of the best-fit cosine curve.

During the lights-off time (2000 to 0700 hours), physiological data were continuously recorded on a Grass Model 78 polygraph. The resulting paper sleep record, containing three channels of EEG and one channel of each EOG, EMG, and EKG was hand scored to provide measures of the sleep variables previously obtained in pigtail infants (see Reite & Short, 1978, for details). Sleep variables scored included: sleep latency in minutes, REM latency in minutes, minutes of each sleep stage (Drowsy, Stage 2, Stage 3 and 4, REM), number of REM periods, mean inter-REM interval, mean REM period length, and total sleep time in minutes.

Behavioral Data Collection and Analysis

During both baseline and separation, the animals were observed for six 5-min periods daily. Observation times were randomly distributed in morning (0900–1130) and afternoon hours (1330–1600) to minimize time-locking to circadian rhythms. Behavioral observations were made by trained observers through a one-way window using a behavioral taxonomy for the bonnet macaque initially described by Kaufman and Rosenblum (1966) and modified by Reite and Short (1980). All behavioral data were collected on-line and subsequently analyzed to provide scores for all variables listed in Table 1. Both frequency and duration behaviors were considered. Frequency behaviors are reported in count/1,000 sec; duration behaviors are reported in percent total time observed (%TTO).

The activity variable was obtained during each observation session. For this purpose, the living pen was divided into 20 approximately equal-sized cubes, the borders of which were some prominent physical structure (a shelf or a pipe) in the area. Every time the infant was observed to cross the boundary of a new cube, a hand-held counter was incremented by 1 count. The number of counts accrued at the end of the observation session was the activity for that session.

Statistical Analysis

Repeated-measures ANOVA was used to test whether behavioral and physiological variables differed significantly between baseline, the first day of separation (SD1), and separation days 2–4 (S2–4). For all significant ANOVAs, post hoc comparisons were made using the Newman-Keuls test.

TABLE 1

A List of Behavioral Variables

Behavior	Description
Activity	Measure of activity described in text
Contact	%TTO spent in physical contact with other animals
Coo vocalizations	Count/1,000 sec of a long tonal vocalization, usually considered a distress call
Drinking	%TTO spent drinking
Eating	%TTO spent eating
Exploration	%TTO spent visually or manually exploring inanimate objects
Play	%TTO spent playing alone or with others
Self-directed	%TTO spent mouthing, licking, or grooming self
Slouch	%TTO spent sitting hunched over with the head down while awake—characteristic posture of depression

Results

Physiological Responses to Separation

Heart rate.—Both mean day and night HR changed significantly in response to separation, $F(2,8) = 18.94$, $p < .01$, and $F(2,8) = 11.27$, $p < .01$, respectively (see Fig. 1). Post hoc tests revealed that HR declined significantly from baseline during S2–4. The increase in daytime HR on SD1 was not statistically significant. Heart rate level showed a significant decrease during S2–4, $F(2,8) = 15.39$, $p < .01$; means for baseline, SD1, and S2–4 were 173, 173, and 145, respectively.

Body temperature.—Mean night body temperature also decreased significantly, $F(2,8) = 11.79$, $p < .01$ (see Fig. 1). Nighttime BT was significantly lower on SD1 and during S2–4 than it was during baseline. Body temperature level was significantly changed, $F(2,8) = 10.80$, $p < .01$; mean values for baseline, SD1, and S2–4 were 38.16, 37.85, and 37.88, respectively. Post hoc analysis indicated a significant decrease from the baseline level on SD1 and during S2–4; this pattern was consistent with the change in mean night BT.

Sleep patterns.—There were significant changes in REM-related sleep variables (see Fig. 2). Post hoc analyses indicated that there

This content downloaded from
81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms

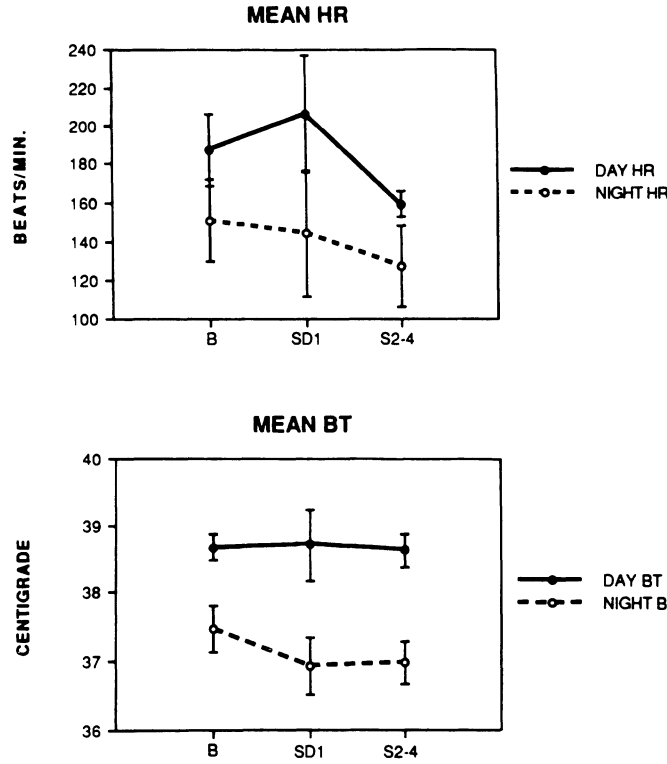


FIG. 1.—Changes in day (solid lines) and night (broken lines) mean heart rate and body temperature. Abbreviations: B = baseline days 1–4; SD1 = first day of separation; S2–4 = days 2–4 of separation.

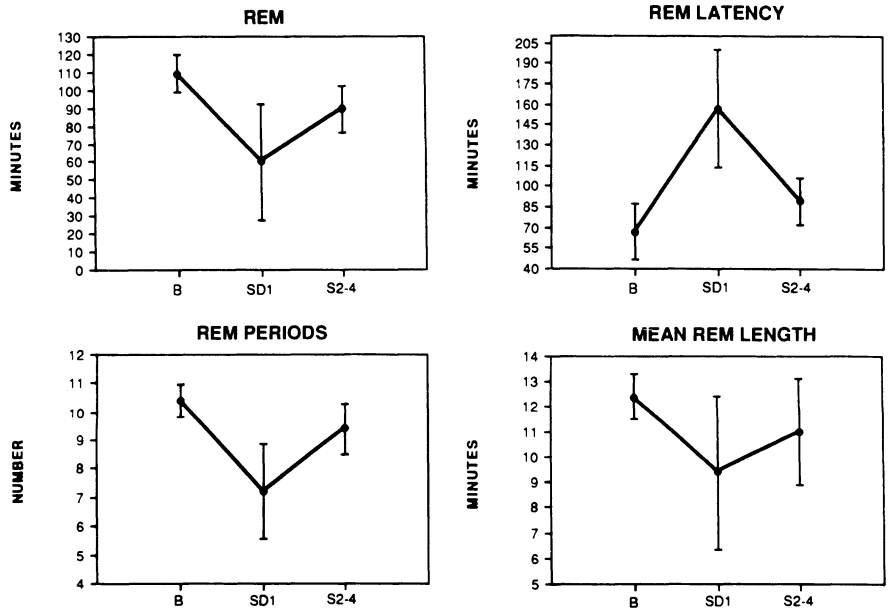


FIG. 2.—Changes in REM-related sleep variables following separation. Abbreviations as in Fig. 1

This content downloaded from
81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms

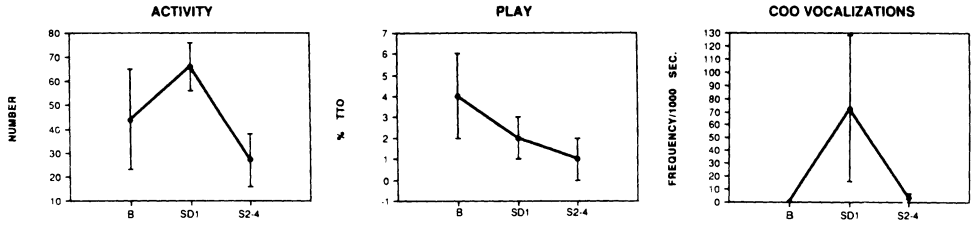


FIG. 3.—Changes in behavior associated with separation. Abbreviations as in Fig. 1

were significantly longer REM latencies, $F(2,8) = 14.13$, $p < .05$, fewer REM periods, $F(2,8) = 8.12$, $p < .05$, and shorter REM periods as determined by mean REM length, $F(2,8) = 7.97$, $p = <.05$, on SD1 the night following separation. Therefore, the total amount of REM sleep was significantly decreased, $F(2,8) = 7.24$, $p < .05$, on SD1. Non-REM sleep was not altered.

Behavioral Responses to Separation

Analysis of the bonnet infants' behavioral responses to separation provides evidence for both the agitation and the depression phase of the separation response (see Fig. 3). There were significant changes in activity, $F(2,8) = 8.74$, $p < .05$, and the number of coo vocalizations, $F(2,8) = 8.27$, $p < .05$, indicative of agitation. Post hoc tests indicated that activity was higher on SD1 than during the rest of separation, and that there were more coo vocalizations on SD1 than during both baseline and the remainder of separation. The changes in the percent total time of play decreased significantly during S2–4, $F(2,8) = 5.01$, $p < .05$. Other behaviors in Table 1 did not change significantly.

Each infant was adopted by and slept with one particular female in the social group. None of these females were lactating. The instantaneous frequency sampling of behavior (every 15 min) that accompanied the physiology data collection revealed that contact with other females (which was the adoptive female) during separation directly paralleled contact with the mother during baseline. This contrasts sharply with the frequency of contact with other females during baseline, which hovered near zero in this measure of behavior (see Fig. 4), $F(6,12) = 38.34$, $p = .01$.

Discussion

Changes in behavior accompanying maternal separation in these five bonnet infants showed evidence of both agitation and depression. The changes in behavior evidencing agitation on the first day of separation were generally similar to previously reported behavior changes following a similar separation experience in pigtail infants (Reite, Short, Seiler, & Pauley, 1981). These changes included increases in activity, cooing, and decreases in social play during the first day of separation. The bonnet infants differed from previously studied pigtail infants, however, in that they showed far less evidence of the subsequent behavioral depression. Play behavior decreased significantly, but there was an absence of slouch behavior, a prominent postural collapse that increases to 20% of the total time observed in separated and depressed pigtail infants.

These bonnet infants also showed a similarity in the nature, if not degree, of physiological changes accompanying separation to those previously described in separated infant pigtail monkeys. There were significant decreases in both day and night HR on S2–4, although the changes were not as pronounced as with pigtails.

The bonnet infants also showed a decrease in nighttime BT during separation. In previous studies of pigtail infants, it was difficult to be certain whether decreases in nocturnal (mean night) BT represented a disturbance in the regulation of BT, or merely the fact that the separated infant was no longer sleeping with a warm mother. The present data tend to support the former explanation since the nocturnal observations of the separated bonnet infants clearly established the fact that they were sleeping in physical contact with another adult female. The loss of a nocturnal heat source as a causative factor in the decreased nocturnal BT can be effectively ruled out.

The changes in nocturnal sleep patterns in these bonnet infants following separation were also similar in terms of their overall pattern and in their predominant emphasis on REM-related sleep variables to the changes previously described in separated pigtail infants. Pigtail infants demonstrate similar increased REM latency, decreased REM time,

This content downloaded from
81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms

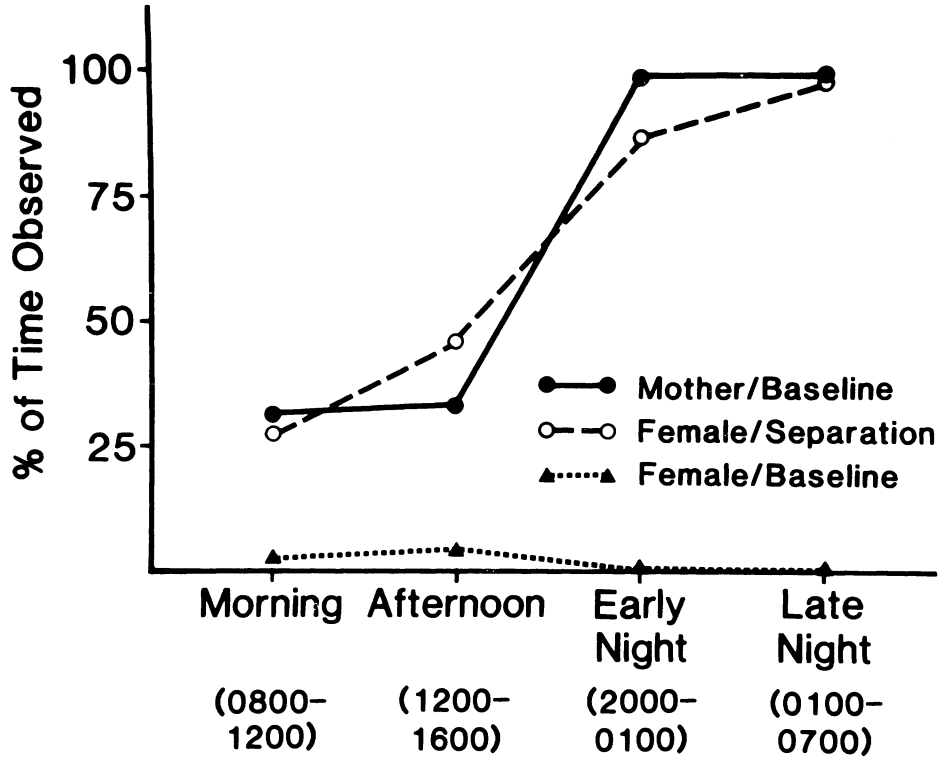


FIG. 4.—Mean percent of 15-min spot observations in contact with mother (solid line) baseline, and other adult females (broken line) during separation, for four time periods during the day. Observations in which infant was in contact with other females during baseline (dotted line) when mother was present approached zero.

and fewer and shorter REM periods the night following separation.

These bonnet infants were adopted by other adult females, generally one particular female in each infant's case. A previous study of pigtail infants demonstrated that separated infants who had social contact with peers in their social group had smaller separation-induced changes in HR than separated infants with no peers present (Caine & Reite, 1981). Thus, the present findings also suggest that the social nature of the separation environment may exert a moderating influence on the infant's response to maternal loss.

An important question is whether it was necessary for the infant bonnet to have developed a previous "attachment" to the adoptive female for the adoption to reduce the response to loss of the mother. Perhaps the simple act of adoption per se, even by a strange female, would have been equally effective. This question cannot be examined systematically with this data base (behavioral data collection during baseline did not differentiate individual females with which the infant made physical contact). Evidence from two case reports, however, suggests that the nature of the prior relation is important.

In one case, a separated 21-week-old pigtail infant was adopted by a childless adult pigtail female within 7 min of its mother being removed from its conspecific group. The infant was familiar with this female but had spent no time in physical contact with her prior to the separation. The infant was held in ventral-ventral contact throughout the first night of separation. Nonetheless, it exhibited a decrease in night BT of 1.4° C, as well as prominent sleep disturbances (Reite, Seiler, & Short, 1978).

In the other case, a bonnet infant separated from its mother and other conspecifics in a mixed-species group was adopted by a pigtail female with whom it was familiar but with whom it had not been in previous contact. The adopted infant demonstrated prominent decreases in HR and BT and developed pronounced behavioral characteristics of depression (Reite, 1985). Thus, in both cases, adoption per se by a familiar female, but one with whom there was no preexisting relationship (which included physical contact), did

This content downloaded from
81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms

not ameliorate the response to separation from the mother. These observations suggest that adoption by an adult female is of limited value if there is no previous contact with the adoptive female.

These observations are interesting in light of a recent study by Brown et al. (1986), who examined the role of social support in influencing the risk of stress-induced (usually loss-related) depression in a prospective study of 400 working-class women. Support from a "core" relationship at the time of the stress was associated with a decreased risk for depression. The presence of crisis support from a "noncore" relationship was not helpful, however, and was in fact associated with a higher rate of depression. These findings are consistent with the hypothesis that a preexisting, special relationship, above and beyond mere familiarity and which might be conceptualized in terms of an "attachment bond," is a prerequisite for social support to have substantial beneficial effects in modifying the response to loss or similar stress.

Attachment, a construct of considerable heuristic value, has emphasized the concept of "regulation" as being of central importance (Hofer, 1984). Field (1985) has proposed that social attachment might generally be construed as psychobiological synchrony between attached individuals that serves to modulate or regulate arousal level and facilitate a state of optimal growth and functioning. Hofer (1984) has discussed at some length the role of social relationships as regulators. The dysregulation in behavior and physiology that accompanies the disruption of attachment is thus conceptualized as the sudden loss of this regulating or modulating influence. The effect of social support from other attachment figures during separation may be to ameliorate the response by providing an alternative source of regulation, or a relationship within which regulation can occur.

Separation studies in both pigtail and bonnet monkeys are consonant with this perspective. The attachment bond with the mother was disrupted, but the adoption by another familiar female with whom a previous relationship may have existed may have served to prevent the more severe disorganization of both behavior and physiology seen in pigtail infants.

We believe these findings suggest that primate models may be useful not only for examining the physiological consequences of separation and loss as they relate to depressive behavioral symptoms, but because primates exhibit complex social systems, the mediating role of social factors in the response to loss may also be modeled with such paradigms.

References

Bakwin, H. (1949). Psychologic aspects of pediatrics. *Journal of Pediatrics*, **35**, 512–521.

Bowlby, J. (1960). Grief and mourning in infancy and early childhood. *Psychoanalytic Study of the Child*, **15**, 9–52.

Brown, G. W., Andrews, B., Harris, T., Adler, Z., & Bridge, L. (1986). Social support, self-esteem, and depression. *Psychological Medicine*, **16**, 813–831.

Brown, G. W., & Harris, T. (1978). *Social origins of depression*. New York: Free Press.

Caine, N., & Reite, M. (1981). The effect of peer contact upon physiological response to maternal separation. *American Journal of Primatology*, **1**, 271–276.

Cassel, J. (1976). The contribution of the social environment to host resistance. *American Journal of Epidemiology*, **104**, 107–123.

Cobb, S. (1976). Social support as a moderator of life stress. *Psychosomatic Medicine*, **38**, 300–314.

Dohrenwend, B. P., & Dohrenwend, B. S. (1974). *Stressful life events: Their nature and effects*. New York: Wiley.

Field, T. (1985). Attachment as psychobiological attunement: Being on the same wavelength. In M. Reite & T. Field (Eds.), *The psychobiology of attachment* (pp. 415–454). New York: Academic Press.

Halberg, F., Tong, Y. L., & Johnson, E. A. (1967). Circadian system phase—an aspect of temporal morphology; procedures and illustrative examples. In H. Von Mayerbach (Ed.), *The cellular aspects of biorhythms* (pp. 20–48). New York: Springer-Verlag.

Hinde, R. A., Spencer-Booth, Y., & Bruce, M. (1966). Effects of 6-day maternal deprivation on rhesus monkey infants. *Nature*, **210**, 1021–1023.

Hofer, M. (1984). Relationships as regulators: A psychobiologic perspective on bereavement. *Psychosomatic Medicine*, **46**, 183–197.

Kalin, N. H., & Carnes, M. (1984). Biological correlates of attachment bond disruption in humans and non human primates. *Progress in Neurology, Psychopharmacology and Biological Psychiatry*, **8**, 459–469.

Kaufman, I. C., & Rosenblum, L. A. (1966). A behavioral taxonomy for Macaca nemestrina and Macaca radiata: Based on longitudinal observation of family groups in the laboratory. *Primates*, **7**, 206–258.

Kaufman, I. C., & Rosenblum, L. A. (1967). The

This content downloaded from
81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms

reaction to separation in infant monkeys: Anaclitic depression and conservation-withdrawal. *Psychosomatic Medicine*, **29**, 649–675.

Laudenslager, M. L., & Reite, M. (1984). Losses and separations: Immunological consequences and health implications. In P. Shaver (Ed.), *Reviews in personality and social psychology* (Vol. **5**, pp. 285–312). Beverly Hills, CA: Sage.

Monroe, S. M., Bromet, E. J., Connell, M. M., & Steiner, S. C. (1986). Social support, life events, and depressive symptoms: A 1-year prospective study. *Journal of Consulting Clinical Psychology*, **54**, 424–431.

Myers, J. K., Lindenthal, J. J., & Pepper, M. P. (1972). Life events and mental status: A longitudinal study. *Health and Social Behavior*, **13**, 398–406.

Pauley, J. D., & Reite, M. L. (1981). A microminiature hybrid multi-channel implantable biotelemetry system. *Biotelemetry and Patient Monitoring*, **8**, 163–172.

Paykel, E. S., Myers, J. K., Dienelt, M. N., Klerman, G. L., Lindenthal, J. J., & Pepper, M. P. (1969). Life events and depression: A controlled study. *Archives of General Psychiatry*, **21**, 753–760.

Reite, M. (1985). Implantable biotelemetry and social separation in monkeys. In G. Moberg (Ed.), *Animal stress* (pp. 141–160). Bethesda, MD: American Physiological Society.

Reite, M. L., & Capitanio, J. P. (1985). On the nature of social separation and social attachment. In M. Reite & T. Field (Eds.), *The psychobiology of attachment* (pp. 223–255). New York: Academic Press.

Reite, M., Seiler, C., Crowley, T. J., Hydinger-MacDonald, M., & Short, R. (1982). Circadian rhythm changes following maternal separation in monkeys. *Chronobiologia*, **9**, 1–11.

Reite, M., Seiler, C., & Short, R. (1978). Loss of your mother is more than loss of a mother. *American Journal of Psychiatry*, **135**, 370–371.

Reite, M., & Short, R. (1978). Nocturnal sleep in separated monkey infants. *Archives of General Psychiatry*, **35**, 1247–1253.

Reite, M., & Short, R. (1980). A biobehavioral developmental profile (BDP) for the pigtailed monkey. *Developmental Psychobiology*, **13**, 243–285.

Reite, M., & Short, R. (1981). Circadian rhythms in monkeys: Variability and behavioral correlations. *Physiology and Behavior*, **27**, 663–671.

Reite, M., & Short, R. (1983). Maternal separation studies: Rationale and methodological considerations. In K. A. Miczek (Ed.), *Primate ethopharmacology: Models for neuropsychiatric disorder* (pp. 219–253). New York: Liss.

Reite, M., & Short, R. (1986). Behavior and physiology in young bonnet monkeys. *Developmental Psychobiology*, **19**, 567–579.

Reite, M., Short, R., Kaufman, I. C., Stynes, A. J., & Pauley, J. D. (1978). Heart rate and body temperature in separated monkey infants. *Biological Psychiatry*, **13**, 91–105.

Reite, M., Short, R., Seiler, C., & Pauley, J. D. (1981). Attachment, loss and depression. *Journal of Child Psychology and Psychiatry*, **22**, 141–169.

Reite, M., Walker, S. D., & Pauley, J. D. (1973). Implantation surgery in infant monkeys. *Laboratory Primate Newsletter*, **4**, 1–6.

Rosenblum, L. A., & Kaufman, I. C. (1967). Laboratory observations of early mother-infant relations in pigtail and bonnet macaques. In S. A. Altman (Ed.), *Social communication among primates* (pp. 34–41). Chicago: University of Chicago Press.

Rosenblum, L. A., & Kaufman, I. C. (1968). Variations in infant development and response to maternal loss in monkeys. *American Journal of Orthopsychiatry*, **38**, 418–426.

Seay, B. M., Hansen, E. W., & Harlow, H. F. (1962). Mother-infant separation in monkeys. *Journal of Child Psychology and Psychiatry*, **3**, 123–132.

Seiler, C., Cullen, J. S., Zimmerman, J., & Reite, M. (1979). Cardiac arrhythmias in infant pigtail monkeys following maternal separation. *Psychophysiology*, **16**, 130–135.

Spitz, R. A. (1946). Anaclitic depression: An inquiry into the genesis of psychiatric conditions in early childhood. *Psychoanalytic Study of the Child*, **2**, 313–342.

This content downloaded from
81.159.84.22 on Thu, 01 Oct 2026 19:21:56 UTC
All use subject to https://about.jstor.org/terms